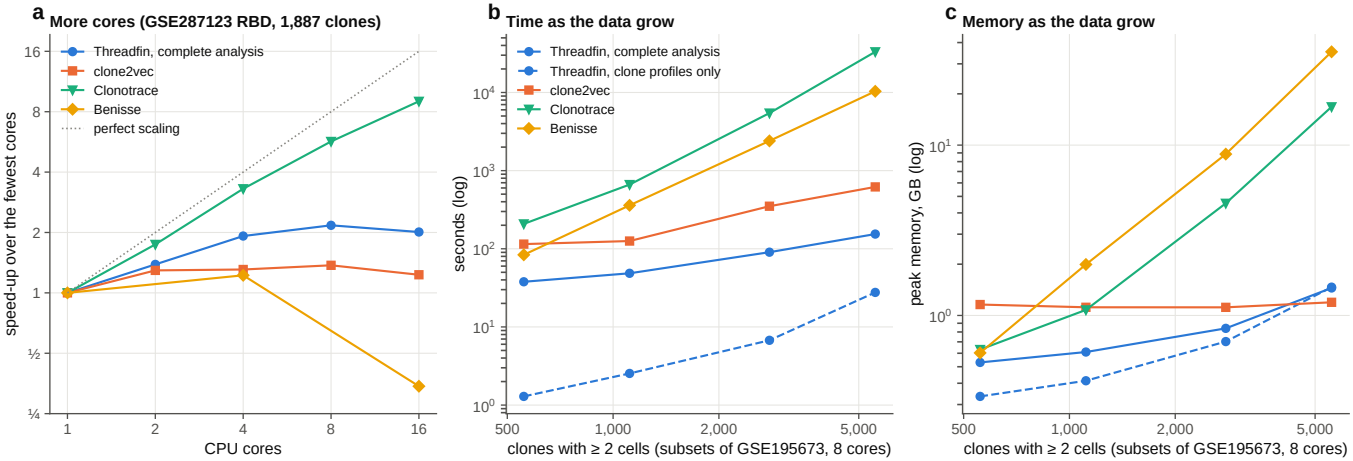
a
More cores (GSE287123 RBD, 1,887 clones)
Threadfin, complete analysis
clone2vec
Clonotrace
Benisse
perfect scaling
speed-up over the fewest cores
16
8
4
2
1
½
¼
1
2
4
8
16
CPU cores
b
Time as the data grow
Threadfin, complete analysis
Threadfin, clone profiles only
clone2vec
Clonotrace
Benisse
seconds (log)
10^4
10^3
10^2
10^1
10^0
500
1,000
2,000
5,000
clones with ≥ 2 cells (subsets of GSE195673, 8 cores)
c
Memory as the data grow
peak memory, GB (log)
10^1
10^0
500
1,000
2,000
5,000
clones with ≥ 2 cells (subsets of GSE195673, 8 cores)